



STREET LIGHT FAULT DETECTION



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ABSTRACT

Streetlights are a very essential part of any road. Faults in these can lead to an increase in energy consumption and cause many accidents. For the minimization of such incidents, it is essential to fix these faulty streetlights at the fastest. The system proposed here is a solution for monitoring the functioning of a street light with addition of an alarming system for ambulances. The technology used behind this is LoRa, which is a wireless communication technology used for long range communication, ESP32, a microcontroller used for short range communication (ESP8266), wireless control and data collection and interfacing with different sensors like photo, current and orientation sensors to detect the faults. It is further embedded with an alarming system for clearance of traffic for ambulances which is done by installing an alarm and connecting with a radio receiver. Using web server programming, the data is displayed on a web page. This project was developed to produce a prototype product of a web-based streetlight fault detecting system that allows the user to detect the faulty street light. The enhancement from the existing systems is that this prototype system allowed the data to be monitored without the use of the internet and created a system for the alert for the clearance of the route for ambulances. This project involves two parts: hardware and software.

Keywords:

LoRa
ESP32
Radio receiver
Ambulance alerting system

1. Introduction

Streetlight systems play an important role in ensuring public safety, energy efficiency, and comfort in urban environments. However, faults in streetlights often increase maintenance costs, lead to energy wastage, and cause illumination problems. This paper presents a streetlight fault detection and alerting system that utilizes current sensors, photosensors, ESP32 microcontrollers, and LoRa communication protocols for real-time fault reporting. The system also integrates an orientation sensor to detect the fallen streetlight poles, which is particularly useful in mountainous regions with sharp curves where continuous illumination is critical for public safety.

Beyond streetlight monitoring, this work also addresses the critical issue of ambulance delays due to traffic. The proposed design employs radio receivers integrated into selected streetlights, which are connected to an alarm system. We will have a website in our headquarters from which we will be able to monitor alarms of each streetlight with the help of LoRa Communications and ESPNOW communications. This mechanism assists in creating an emergency corridor, thereby reducing response times and improving emergency services.

The proposed solution is cost-effective, making it suitable for enhancing streetlight maintenance and energy management in urban environments.

1. Related works

Countries have millions of streetlights. It is important to detect faults in it, as it is put in work for the purpose of providing lighting in surroundings, so if they are not in working condition, they are not serving their basic purpose. Though there are various existing and proposed solutions based on IOTs, wifi based communications, GSM based communications, LoRa communications; many of them face high power consumptions, short ranges, monthly recharges and short period maintenance. Also, because we are implanting microcontrollers in most of those cases, there are large amounts of those that are underutilized, so there will also be a scope of future research on its utilizations.

- In [1] the authors used some sensors like light sensor, photo sensor, Ultrasonic sensor and LDR to detect the fault in the street light and give information about this to the center using GPS technology.
- In [2], [3] the authors used LoRaWAN (Long Range Wide Area Network) to develop a smart solution for detecting faults and automating the response over long distances with very low power consumption. Unlike in GSM and GPS systems, LoRa does not require any recharge with a working range of more than 10-15 km.
- In [4] authors suggest the idea to integrate LoRaWAN and ESPNOW communication technologies. They both work with limited energy but perform critical communication functions effectively. ESP-NOW enables short-range communication between ESP devices, whereas LoRAWAN provides long-distance transmission, The combination of these two technologies offers a cost effective and efficient solution for communication.
- In [5],[6] and [7], the authors propose a system that combines motion sensors and LDR (Light Dependent Resistor) sensors to control street lighting based on nearby movement and ambient light conditions. They also recommend an odd/even lighting scheme, where alternate lights are switched on, further optimizing energy consumption. This method helps to improve system efficiency, reduces energy usage and introduces the concept of an IoT-based dashboard for remote monitoring and management of street lights.
- In [8], 2 Author implemented system that uses traffic flow information worked on doppler effect to detect it and that system was implemented in Luyang Avenue, Lucheng City, Shanxi Province, China in May 2019, which reduced electricity consumption in that region by ~18%.
- [9], [10], there is research on daily motionless activities of a human using Gyroscope, Accelerometer and magnetometer. The Gyroscope measures how fast an object rotates around the X, Y, Z axis. Accelerometer measures linear acceleration in the X, Y, Z axis. Magnetometer measures the direction of Earth's magnetic field. Since these sensors detect change in orientation of the street light pole (eg. Check if the pole has fallen) and poles fall mostly in hilly areas and very few fall in urban areas so

that sensor can be made attachable so that it can be deployed where it is required.

- In [11] the author proposed a smart ambulance system that computes an emergency route via Google maps APIs, the system then shares information with mobile users. The main base of this work was to inform all the other drivers on the same route as the ambulance, instead of depending on the vehicle-to-vehicle or traffic management system. The author noted that if drivers do not keep his mobile app open then he would miss notification. Then the authors also suggested that future alerting systems should not rely completely on phone usage. This suggestion is important for our work, as it motivates us to use the microcontroller of our street light fault detection system for serving the purpose.
- In [12] the authors proposed an idea to solve the ambulance problem statement using the Zigbee module and when it came near the receiver then an alarm system turned ON to give information about the ambulance and it will work with the algorithm of master slave and master control.

2. Existing Products

- iSSL by ITHENA, INDIA
 - Functionality: It uses NB-IOT based real-time fault alerts to improve energy efficiency, reduce maintenance costs and enhance overall light management. It includes adaptive brightness based on vehicular or human presence, and an automated schedule for turning the light on and off.
 - Limitations: requires regular recharging, individual lamp monitoring is required, poor NB-IOT coverage in rural areas.
 - Similarity with our solution: LoRa/LoRaWAN for long-range low-power communication, adaptive brightness system
- CityTouch Connect Cabinet Controller by Philips Signify
 - Functionality: It monitors a group of streetlights. It measures voltage, current, power factor and energy consumption and detects the unauthorized opening of cabinet door. Uses web-based dashboard for real-time status and cellular network (like 2G, 3G and 4G).
 - Limitations: it cannot identify the exact location of a fault within a group of streetlights, poor network availability in rural areas and the recurring cost of cellular network.
 - Similarity with our solution: real-time status with the help of web-based dashboard, voltage and current monitoring
- CityLight by Motwane, India (Nashik, Maharashtra)
 - Functionality: An integrated product that combines the streetlight with the dimmer. IoT-based automated, real-time monitoring and control of individual streetlights. Uses HPSV lights and adaptive brightness system to save power.
 - Limitations: requires replacement of existing streetlights, it is dependent on continuous network connectivity, requires higher gateway cost.
 - Similarity with our solution: adaptive brightness system, individual monitoring of lights, LoRa gateway to monitor/control light groups
- Waspote Plug & Sense by Libelium (Spain)
 - Functionality: This device can integrate multiple sensors, allowing it to be used for various tasks, one of which is smart streetlighting. It is a long range, low-power and retrofit-friendly device. It supports multiple communication protocols (LoRaWAN, Wi-Fi, Sigfox).
 - Limitations: cannot identify fault at pole-level.
 - Similarity with our product: Fault detection, adaptive brightness based on motion detection and scheduling of switching of light

- sylCloud by SYL Technologies sylCloud
 - Functionality: It is a LoRa-based smart streetlight system. It uses map-based UI, dynamic dimming, and current/voltage monitoring. It can detect overheating, voltage/current supply issues, power supply failures,
 - Limitations: it requires continuous power and dense LoRa gateway placement in cities.
 - Similarity with our solution: uses LoRa for communication, assigns a unique ID to each device for easy identification, and supports current, voltage, and light monitoring for fault detection
- CCMS by TRUEiSENSE (OCTIOT, India)
 - Functionality: It is an IoT-based central control and monitoring system. It works on adaptive brightness based on motion detection, an automated alert system and control over individual streetlights.
 - Limitations: it is dependent on network connectivity
 - Similarity with our solution: Automated alerts and control over individual streetlights
- Smart Street Light Monitoring & Control by Neona Embedded Labz (India)
 - Functionality: Cloud-based IoT system for managing and controlling groups of streetlights using company's Data Connector Unit (DCU). The system uses scheduling to automatically turn lights on and off. It can store data for 45-60 days in case of network failure.
 - Limitations: Scales well for highways, is dependent on network connectivity, does not provide real time information during network failures and involves higher maintenance complexity.
 - Similarity with our solution: Streetlight fault Analysis
- inteliLIGHT by Flashnet / LORIOT (Europe) – Hafnarfjörður, Iceland
 - Functionality: The product is based on adaptive LoRaWAN controllers with Central Management System (CMS), enabling real-time alerts. inteliLIGHT FRCM cabinet controllers and Zhaga-based controllers are used in it.
 - Limitations: group-level control rather than per-lamp sensor granularity, requires reliable LoRaWAN coverage, and involves higher cost
 - Similarity with our solution: it uses an AC–DC converter and operates in always-On Operation • SmartSky / Smartlight (LoRaWAN) by LOT
- EC (Turkey)
 - Functionality: This system uses Light Control Unit (LCUs) are installed on each streetlight and communicate with the help of LoRaWAN controllers and DALI2(for sending data to MCU), it features a map-based interface for control, inbuilt GPS and App/web access.
 - Limitations: Dependent on network connectivity and involves high costs
 - Similarity with our solution: includes sensors, automated dimming, on/off scheduling of lights, and web-based monitoring
- ALC-P1 by Atreyo
 - Functionality: This system uses long-range, radio-frequency-based devices (LoRa, etc.) for transmission of signals. It works on adaptive brightness system with automated on/off of streetlights and measures voltage, current and power. It supports Over the Air (OTA) for updating the fault detection logic.
 - Limitations: We cannot get the pole level problem by this model
 - Similarity with our solution: LoRa-based communication, adaptive brightness and current and voltage measurement

3. System Overview

The system is developed for the purpose of monitoring the streetlights and providing information regarding any fault in it. It is both hardware and software-based product.

- Objectives:
 1. Identify tripping or falling of the Miniature Circuit Breaker (MCB) and generate an alert when the circuit is not completed
 2. Detect tilting or falling of street light pole and generating an alert for maintenance or safety action
 3. Detect the condition where current is flowing through the circuit but the LED is off
 4. Detect abnormal flickering of street lights
 5. Ambulance alerting system that communicate to assist emergency vehicle movements
 6. Automatically dim/off street light when no motion is detected
- Advantages:
 1. Real time fault detection for fast maintenance or safety action
 2. Fast movement of emergency vehicles
 3. No internet connection or wi-fi is required for communication
 4. Continuous monitoring and fault detection in streetlight with their accurate location
 5. No sim card is required, hence prevents the recharge cost
 6. Much cheaper and more accurate compared to other models
 7. Works even during the power cut with the help of battery
 8. Charges the battery with the current in streetlight, hence there is no need to change battery

4. Hardware Used

- 1) ESP8266 Micro controller: Each branch node uses the ESP8266 microcontroller[14], which communicates with the ESP32 head node through ESP-NOW protocol or with other branch node through ESP-NOW protocol according to its positions. This low-latency, thus peers-to-peers communication will reduce cost of external communication devices providing a local communication network. The ESP8266 collects sensor data, processes it and does as instructed. The main benefit of this is it has 11-usable GPIO pins which is enough for this system.
- 2) ESP32 Microcontroller: Each head node uses the ESP32 microcontroller[13], which communicates with the headquarters with the help of LoRa communications(head node). ESP32 has 30-usable GPIO pins allowing us to connect all sensors and Lora module. With its dual-core processor, integrated Wi-Fi, it is a good option since we require more pins.
- 3) ACS712 Current Sensor: The ACS712 5A Hall-effect current sensor[15] measures the real-time current consumption of connected loads or devices. Its high accuracy and galvanic isolation make it a good option for low-voltage IoT systems. In this system its high sensitivity is very useful.
- 4) MPU6050 Accelerometer and Gyroscope: The MPU6050 sensor[16] combines a 3-axis accelerometer and a 3-axis gyroscope to measure motion, tilt, and orientation. Data from this sensor allow the system to detect instant movements, enabling predictive maintenance or motion-based control applications. The sensor communicates with the microcontroller via I²C interface for efficient data transfer, which is useful for pole falling detection in this system.
- 5) BH1750 Light Intensity Sensor: The BH1750 sensor[17,18] is a digital light sensor that helps in detecting the amount of light intensity falling on the sensor in lux units. This sensor is useful in detecting the state of light (ON/OFF).

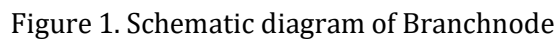
- 6) TP4056 Battery Backup System: The TP4056 module[21] is employed for charging and managing the lithium ion batteries used in the system. It provides overcharge, over-discharge, and short-circuit protection, ensuring safe and reliable operation of the power supply. The module takes care of battery charging, so there is no need of changing batteries.
- 7) 5V Relay Module: A 5V relay module[22] connected to the branch nodes to switch electrical loads based on sensor inputs or commands from the head node. The relay offers galvanic isolation between the low-voltage control circuitry and the high-voltage load. This device allows the microcontroller to do switching in circuits.
- 8) 220V AC Electronic Dimmer (E-Dimmer): The 220V ACE-Dimmer[23] allows to control the brightness of LEDs based on instructions received from the microcontroller. This helps reduce power consumption of street lights, especially in remote areas.
- 9) LoRa SX1278 433 MHz Transceiver Module: The LoRa SX1278 transceiver[19], operating at 433 MHz, facilitates long-range, low-power wireless communication between the head node and headquarters. Its low energy consumption and extended range make it ideal for distributed IoT networks that demand robust and scalable connectivity.
- 10) HC-SR04 Ultrasonic Sensor: The HC-SR04 sensor[20] is used for distance measurement, allowing us to measure change in distance for detecting motion and also distance itself, for measuring stable objects.

5. Design Overview

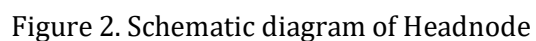
The proposed system uses ESP32 microcontroller in the headnode module, which contains Lora Communication module allowing it to communicate with farther devices and also supports ESP-NOW protocol allowing low-latency communications with branch nodes. Displayed Fig. 3 is the block diagram of the proposed system. A number of sensors are integrated by the branch nodes to monitor different physical properties. While the MPU6050 offers accelerometer and gyroscope data for motion and orientation detection, the ACS712 5A current sensor detects electrical current for the monitoring of electricity. The BH1750 sensor measures ambient light intensity, while the HC-SR04 ultrasonic sensor helps in measuring human traffic. A buzzer for alerts, a 5V relay module for electrical load switching or a servo motor for mechanical control and an electronic dimmer (E-Dimmer) for changeable brightness control are some of the devices used in actuation.

Two systems have been designed for different purposes

- i. Head nodes
- ii. Branch nodes



Head nodes: It is installed at particular streetlights. It receives the data from branch nodes and transfer it to control center using LoRa.



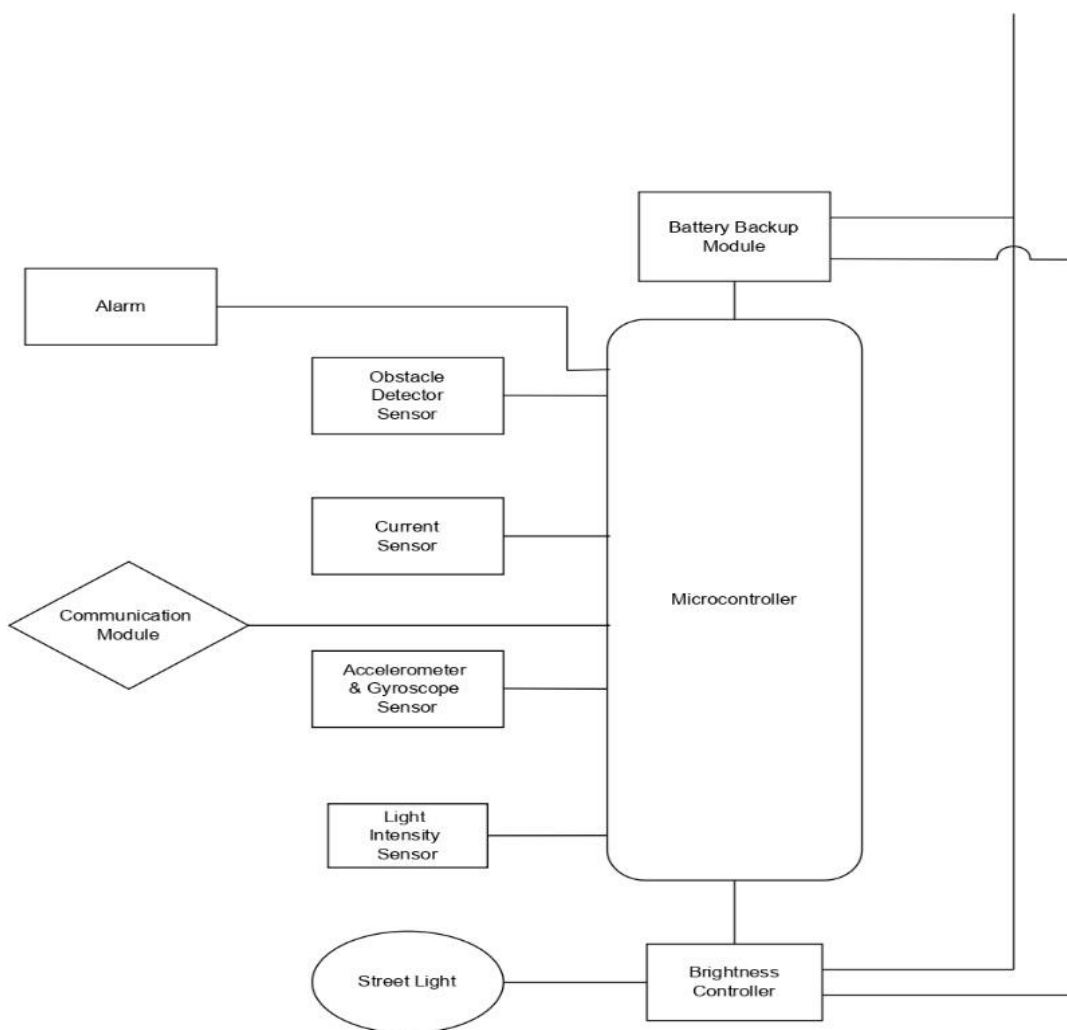


Figure 3. Block Diagram

6. Working Principle

First branch node will start detecting and calibrating sensors, it will take readings of MPU6050 sensor, current sensor calibration, and detect weather all sensors including BH1750 is connected or not, if any of these is not connected it will generate fault message with all faults mentioned, the code has been written in a way such that it generates all such fault messages(some of faults are mentioned ahead) at duration of 30 seconds, till then it will just keep recording all readings and noticing all faults. The fault of fallen pole will be detected by change in readings of MPU6050, since it should be detected instantaneously as if the pole falls it can damage the system, so system should not wait for 30s if fault is of pole fallen, it should instantly generate message in this case, which is a feature of the code. Now it will detect if for some reason MCB has fallen and current is not going to LED light, then current sensor will not get detected by current sensor. Now if current is flowing but light has not turned on then in that case a photosensor will detect fault and generate a message but current sensor won't since current is flowing. the feature of detection of flickering of LED light with the help of a photosensor has been added, through applying logic of continuous high and low readings of photosensor. Also in the case where current sensor is not detecting current and generating false faults messages, it has also been taken care in the code. All the system will be backed up by battery in case of power failure, till then it will work on adapter + cable power. The functionality mentioned above will be for both: Branchnode and Headnode. In case of Headnode(Fig. 5), the extra functionality of LoRa communication will be there, every branchnode will send fault message to Headnode by the help of espnow, and then later on Headnode will transfer all these received fault messages to headquarter with the help of LoRa module installed in it, also it will have its own fault detecting system for own pole too. Now for clearing traffic for Ambulance, a mesh of network has been developed in which each head node has its own head node Id, if that head node id matches with the head node id transmitted then that head node will know that this alert message is either for its own or for its branchnodes. Now for reaching a particular targeted node, all head nodes will also have another id for particularly targeting it which will be in the same order with other branch nodes, those branch nodes will have its own branch node Ids only. Now whenever we transmit message of ALERT, first all head node will check if message Id of headnode in transmitted matches with its id or not, if it matches it will check if its branchnode id matches or not, if it doesn't matches then it will transmit message to the 1st branchnode of that branch, that will check if id matches, if it will then ALERT message will successfully received by targeted node and buzzer will go on, if it doesn't then it will transmit the same message to next node with the help of espnow, and this thing will continue until message die out, message will die out when it will reach to correct node. A feature of all node messaging is also developed, which will send a message alert to all nodes of a particular headnode(including headnode since it will also have its own id+system). A feature of dynamic timing for buzzer being on has been added by which we will be able transmit the amount of time for buzzer to go on along with the message.

7. Manufacturing

1) PCB

The PCB of this system was designed considering the electrical requirements of all the electrical components. Schematic diagram was designed with correct pin mapping for all the electrical components. Adequate spacing between all the components has been given to reduce attenuation and interference. ESP has been placed centrally for easy connectivity with other components and photo sensor was placed near the enclosure to receive unrestricted light. Soldering has been done manually for prototype development. After assembly, the PCB was tested to endure the proper connectivity.

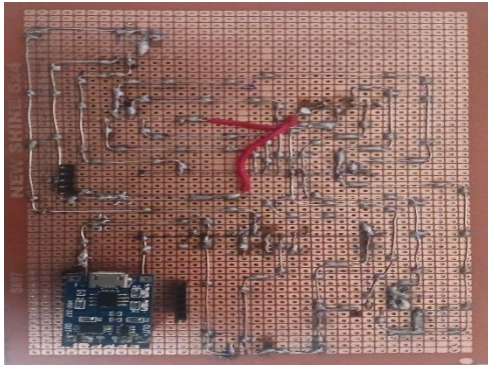


Figure 4. PCB design

2) Boxing

The boxing of the streetlight fault detection unit was made up of 4 mm thick acrylic sheet as it offers good strength, electrical insulation, weather resistance and transparency for sensors that require optical exposure. This enclosure protects all the components from rain, dust and damage. The design of box was created in SolidWorks, the top face of the box is left transparent so that light fall directly over the photosensor so that it can detect the intensity of light without any resistance. The 2D profiles of each panel were exported as DXF files and laser cutting of acrylic was done to ensure clean edges and accurate dimensions, these panels were then joined by L-brackets. Finally, the PCB was mounted inside this box and was tested for the better functioning.

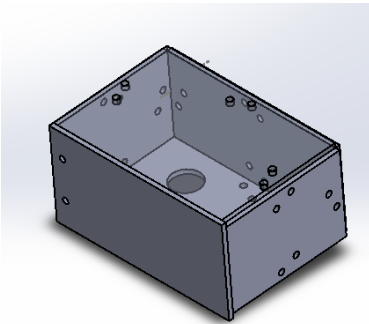


Figure 5. CAD design of enclosure

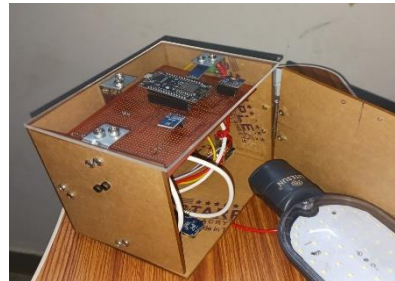


Figure 6. Street light fault detection system

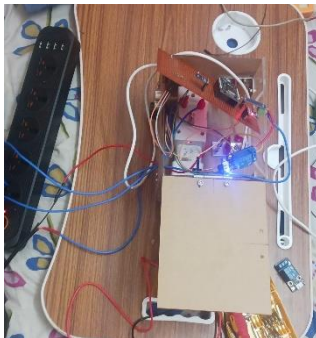


Figure 7. Assembling enclosure and PCB

8. Testing

The complete system underwent a structured testing process to ensure that it responded correctly to all fault scenarios.

1. When power supply was turned off, a “No current detected” warning was received along with the ID of that system.
2. When the system was moved, a “Pole fall detected” warning was generated along with system ID.
3. When power supply was turned on but the light was removed away from the system(condition of LED not working), “Low Light” fault was detected along with system ID.
4. When light was flickered over the system, “Flickering” fault was received along with system ID.
5. When the system received the message for ambulance, the buzzer started ringing. The operator can also manually decide the time for which buzzer has to ring.

Objectives 1, 2, 3, 4 and 5 were successfully achieved.

Objective 6 was not achieved, as the voltage was not enough to make adjustments in the intensity of light. We tried to achieve it by using servo+dimmer, relay module switchoff and ultrasonic sensor by was not able to control the intensity.

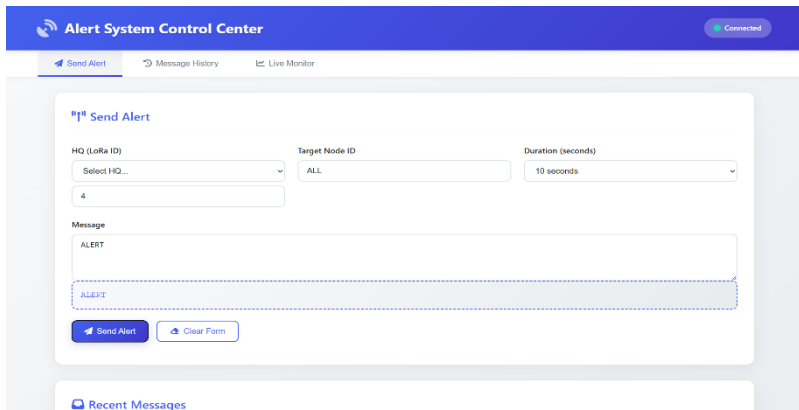


Figure 8. Manual Alert Control

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